1 OFF-GRID POWER solar -> LiFePO4 -> 12 V / 5 V

U1
MPPT charger, 4S LiFePO4, CV 14.4 V
PV1
30 W flex
PV+
BAT+
GND
BT1
4S BMS, 12.8 V 6 Ah LiFePO4 + BMS
CHG+
P+
GND
F1
10 A
+12V
U2
LM2596-5.0
VIN
OUT
ON/OFF
GND
FB
L1
33 uH
SS34 D1
C1
680 uF
C2
220 uF
+12V
+5V
GND

2 SENSE graphite E-skin half-bridge + ratiometric sigma-delta front end

U7
SHT31
VDD
SCL
SDA
GND
+3V3
R7
4.7k
R8
4.7k
I2C_SCL
I2C_SDA
R1
100k (active)
R2
100k (dummy)
BRIDGE_MID
NTC_IN
100k 0.1%(ref)
100k NTC
INT
100k 0.1%(fb)
Q1
2N7002
D
G
S
10R3
R4
100k
C3
10 nF C0G

U3 Sipeed Tang Nano 9K (Gowin GW1NR-9)

5V
3V3_OUT
GND
Phase / chop sequencer (EN 20 ms)
Σ-Δ ratiometric ADC (LVDS comparator)
CIC³ + MA-5 (50 Hz null)
INT8 TCN (1 MAC, ~1.7k params)
CUSUM + consensus gate
LVDS_A+
LVDS_B-
LVDS_A-
LVDS_B+
SD_FB
EN_BRIDGE
UART_TX
UART_RX
TAMPER_IRQ
PELTIER_REQ
Ratiometric: bridge, NTC and 1-bit feedback
GPIOs share VCCIO -> supply drift cancels.
No precision reference IC is required.
Excitation: 20 ms each 1 s (2% duty).
approximately 5 uW average.
Targets: >=13 ENOB @ 10 Hz; INT8 error <1%
vs FP32; latency <=2 s; <=1 false alarm per
1000 device-hours.

4 COOL Peltier lid driver (backup top-up only)

U8
BTS7960 module (IBT-2), 43 A half-bridge
RPWM
B+
M+
M-
B-
R_EN
L_EN
VCC
LPWM
GND
TEC1
TEC1-12706 (12 V, 6 A max)
C5
1000 uF
TEC_EN
PELTIER_PWM
R9
10k TEC_EN pull-down
PCM packs hold 4 +/- 2 degC for >= 8 h
TEC fires only if T_int >= 5 degC
PELTIER_REQ -> ESP32 PWM + dual enable
Hot-side NTC and fan tach feed ESP32 inputs:
firmware drops TEC_EN/PWM on over-temp or tach fault.

5 SETTLE ESP32-S3 + 5G/GNSS + smart contract

U4
ESP32-S3
RXD
TXD
GPIO_IRQ
GPIO_REQ
HOT_NTC_ADC
FAN_TACH
PWM_OUT
TX1
RX1
MDM_PWR
TEC_EN
5V
GND
U5
SIM7600-class 5G/LTE + GNSS modem
RXD
TXD
PWRKEY
MAIN
GPS
VBAT
GND
AE1
5G/LTE MAIN
RF
AE2
GNSS (GPS)
RF
C6
1000 uF
TH2
10k NTC hot-side (assumed)
R11
10k fan-tach pull-up
J1
Fan tach input (verify pinout)
R10
10k 1% NTC divider
AGND
TACH
U6
12 V -> 4.0 V buck, 3 A
VIN
VOUT
Assumes 10k NTC; verify beta/value. J1 tach is open-collector.
Firmware must detect missing tach pulses and disable TEC_EN/PWM.
ECDSA-signed events + GPS -> Solidity quality-score contract
Score = 100 - sigma(T_act - T_set)*t - penalty
Payout: > 90 : 100 % | 70-90 : 80 %
<= 70 : 50 % | tamper flag -> review